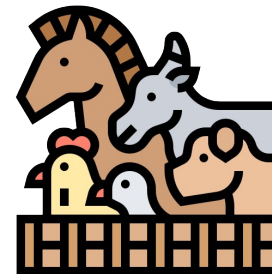
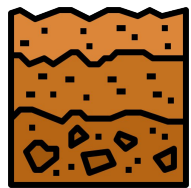


ARC based Middleware – Exchange of domain specific information

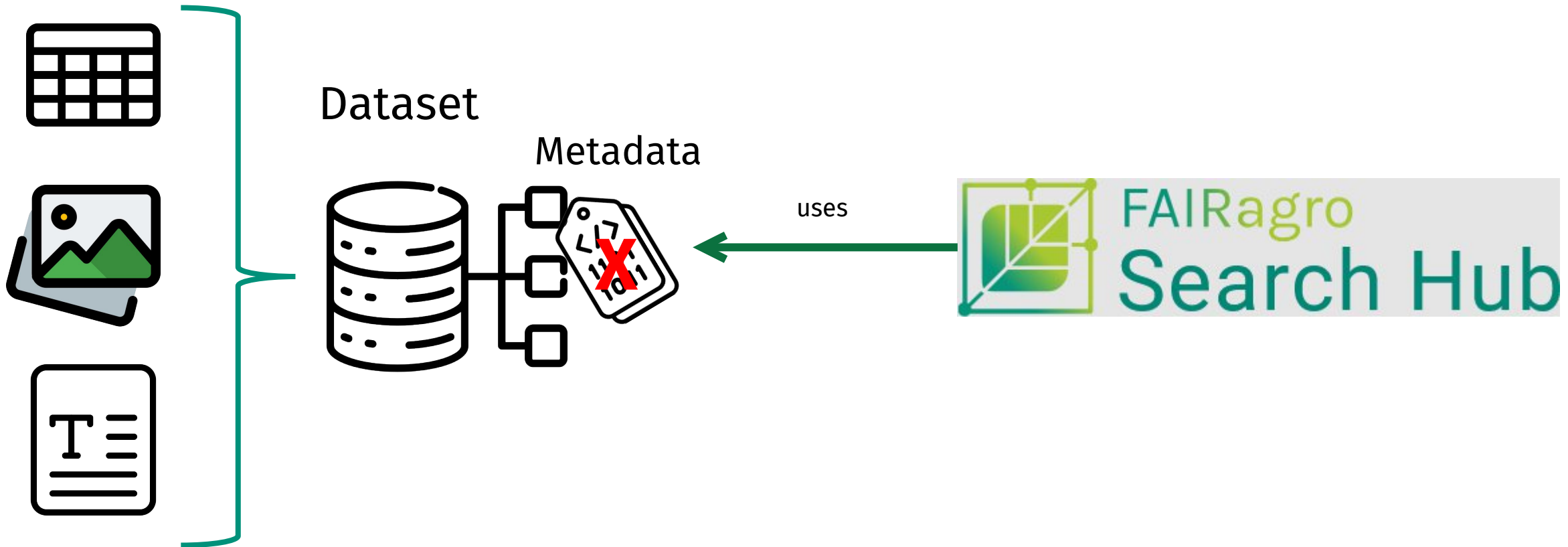
TA3/4 Retreat 2026

Jorge B, Carsten C, Daniel A, Julian S, Gabriel S

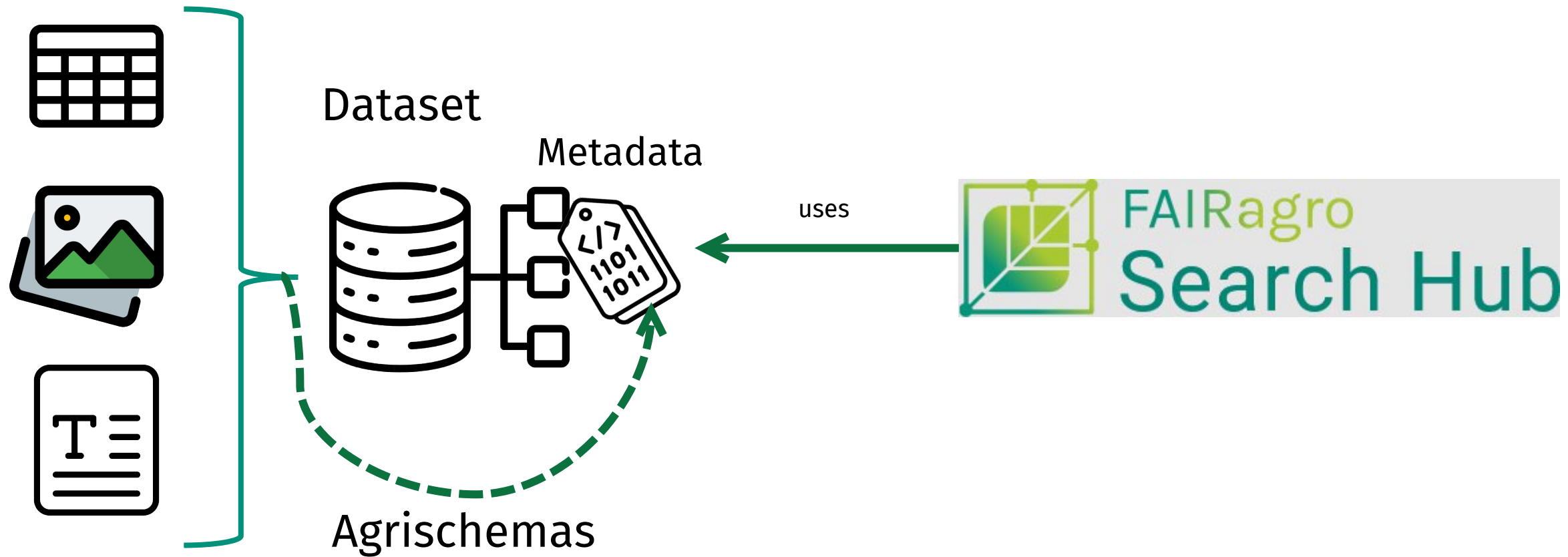
Domain-specific information in FAIRagro RDIs



Using domain-specific information

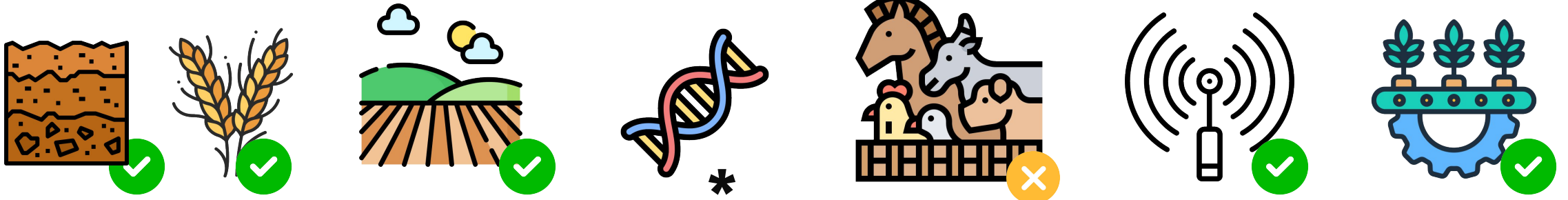


Using domain-specific information



Agrischemas

- Domain-specific information becomes expressible in a Schema.org / Bioschemas conformant way
- Set of classes and properties focused on findability



- Challenge: Handling the variety of ARCs to identify the relevant information and make it available in the FAIRagro Search Hub

*genotype referencable
via Crop class

Relation between Agrischemas and the ARC

Agrischemas Conversion: Starting with the ARC RO-Crate

Project

flyover_redacted

AgDaFAIR / flyover_redacted

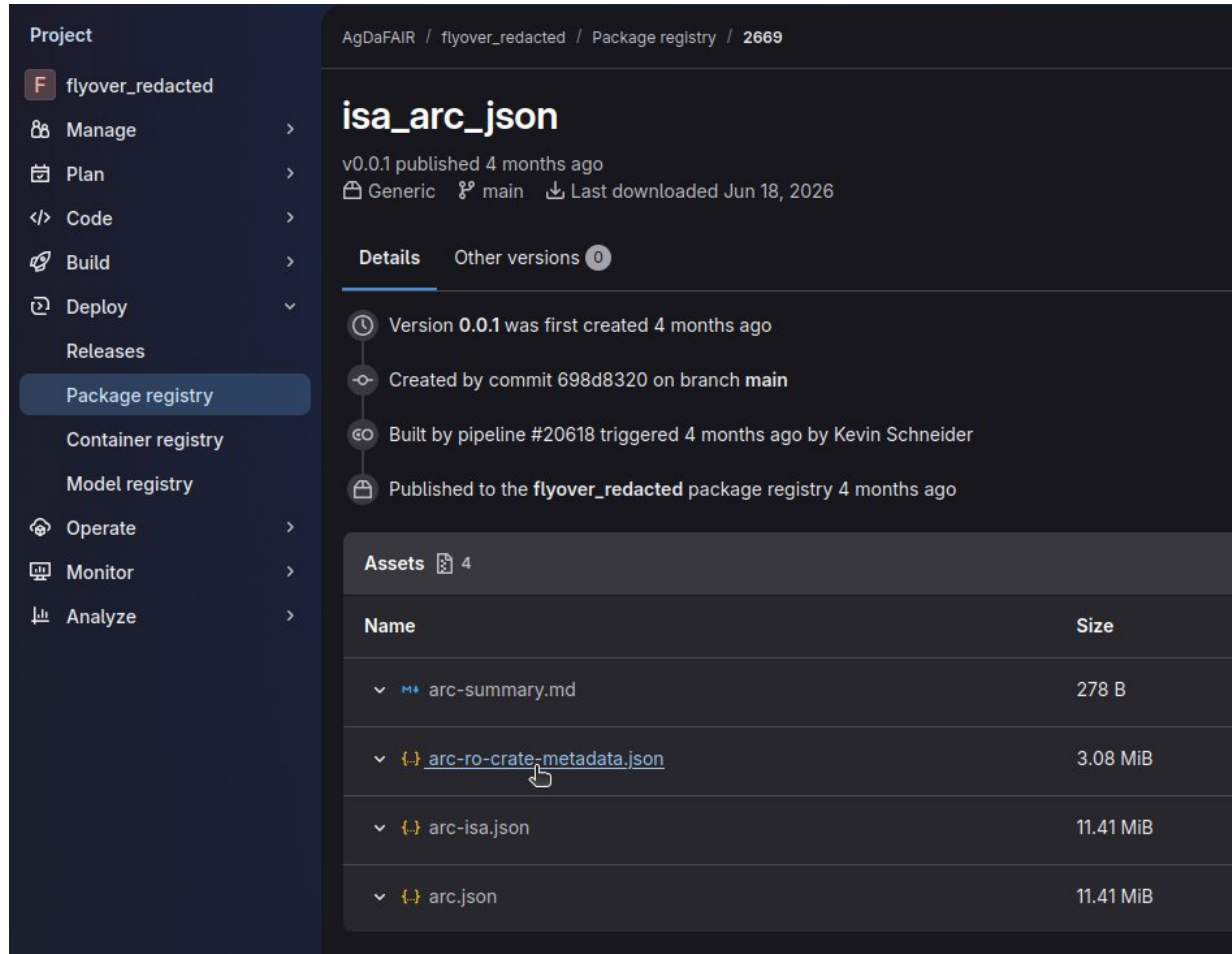
F flyover_redacted

main flyover_redacted Find file Code

add validation
Kevin Schneider authored 4 months ago 698d8320 History

Name	Last commit	Last update
.arc	add validation	4 months ago
assays	init redacted sample arc	4 months ago
runs	init redacted sample arc	4 months ago
studies	init redacted sample arc	4 months ago
workflows	init redacted sample arc	4 months ago
.gitattributes	init redacted sample arc	4 months ago
.gitignore	init redacted sample arc	4 months ago
isa.investigation.xlsx	init redacted sample arc	4 months ago

Agrischemas Conversion: Starting with the ARC RO-Crate



The screenshot displays the FAIRagro Package registry interface. On the left is a dark sidebar with a 'Project' section containing 'flyover_redacted' and a list of navigation items: Manage, Plan, Code, Build, Deploy, Releases, Package registry (highlighted), Container registry, Model registry, Operate, Monitor, and Analyze. The main content area shows the breadcrumb 'AgDaFAIR / flyover_redacted / Package registry / 2669' and the package name 'isa_arc_json'. Below the name, it states 'v0.0.1 published 4 months ago' and 'Generic main Last downloaded Jun 18, 2026'. A 'Details' tab is active, showing a timeline of events: 'Version 0.0.1 was first created 4 months ago', 'Created by commit 698d8320 on branch main', 'Built by pipeline #20618 triggered 4 months ago by Kevin Schneider', and 'Published to the flyover_redacted package registry 4 months ago'. Below this is an 'Assets' section with 4 items listed in a table.

Name	Size
arc-summary.md	278 B
arc-ro-crate-metadata.json	3.08 MiB
arc-isa.json	11.41 MiB
arc.json	11.41 MiB

Examining ARC structure

Goal:

- define how Agrischemas-related objects should be modeled in ARCs
- enable unambiguous transformation

Investigation

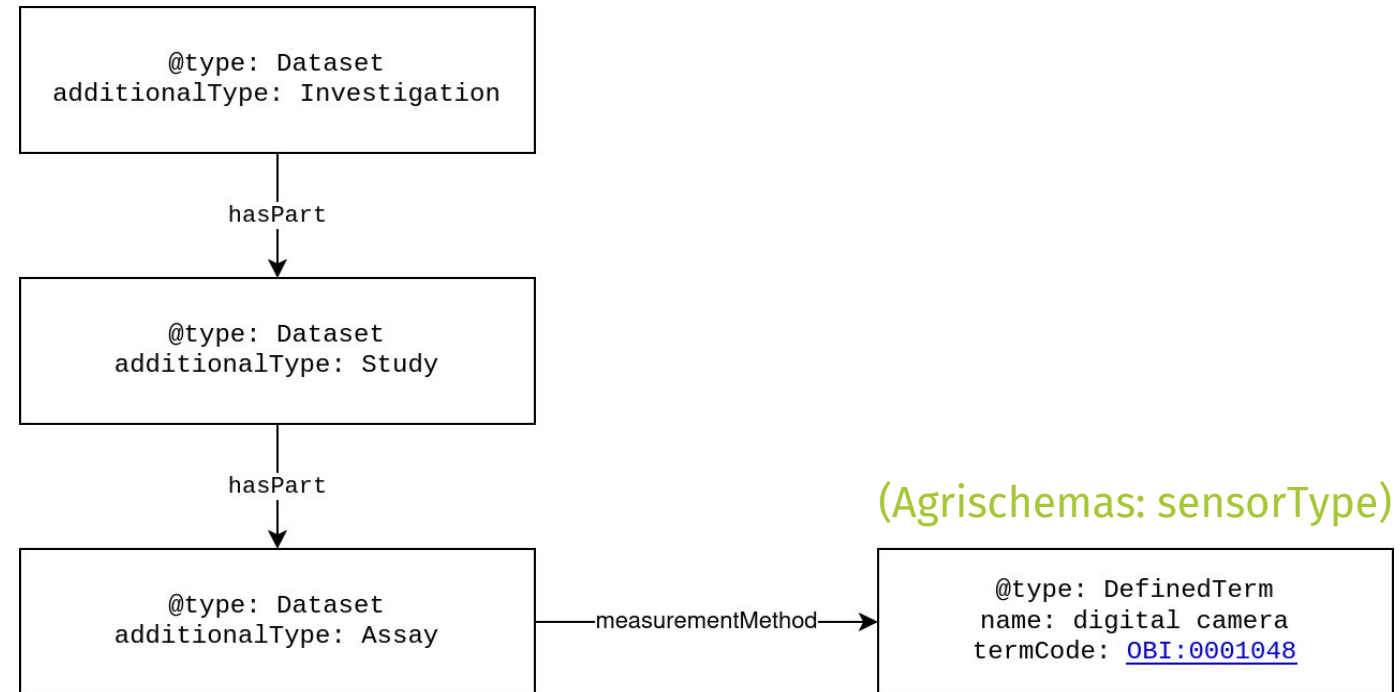
Study

Assay

Examining ARC structure

Goal:

- define how Agrischemas-related objects should be modeled in ARCs
- enable unambiguous transformation

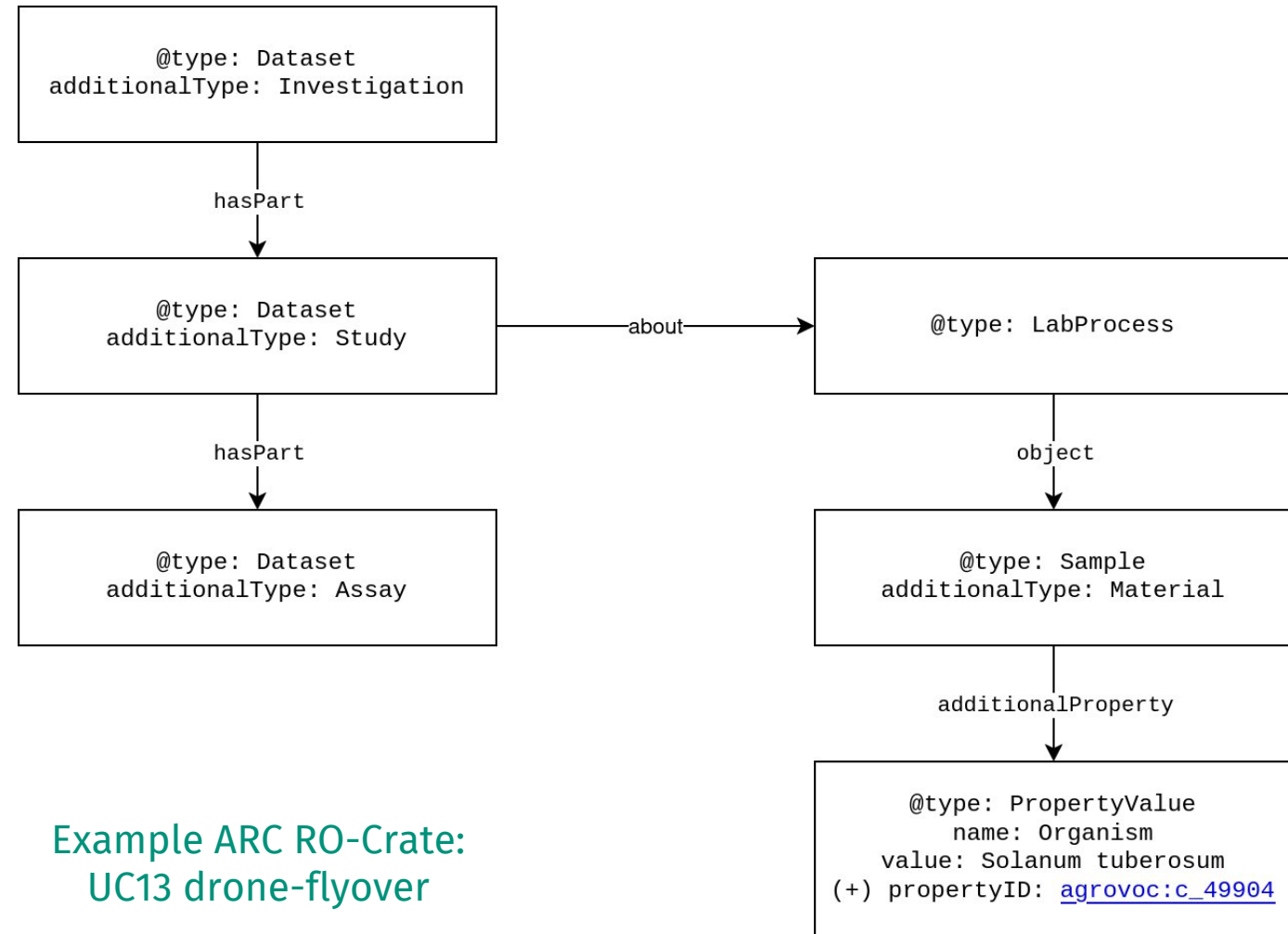


Example ARC RO-Crate:
UC13 drone-flyover

Examining ARC structure

Goal:

- define how Agrischemas-related objects should be modeled in ARCs
- enable unambiguous transformation



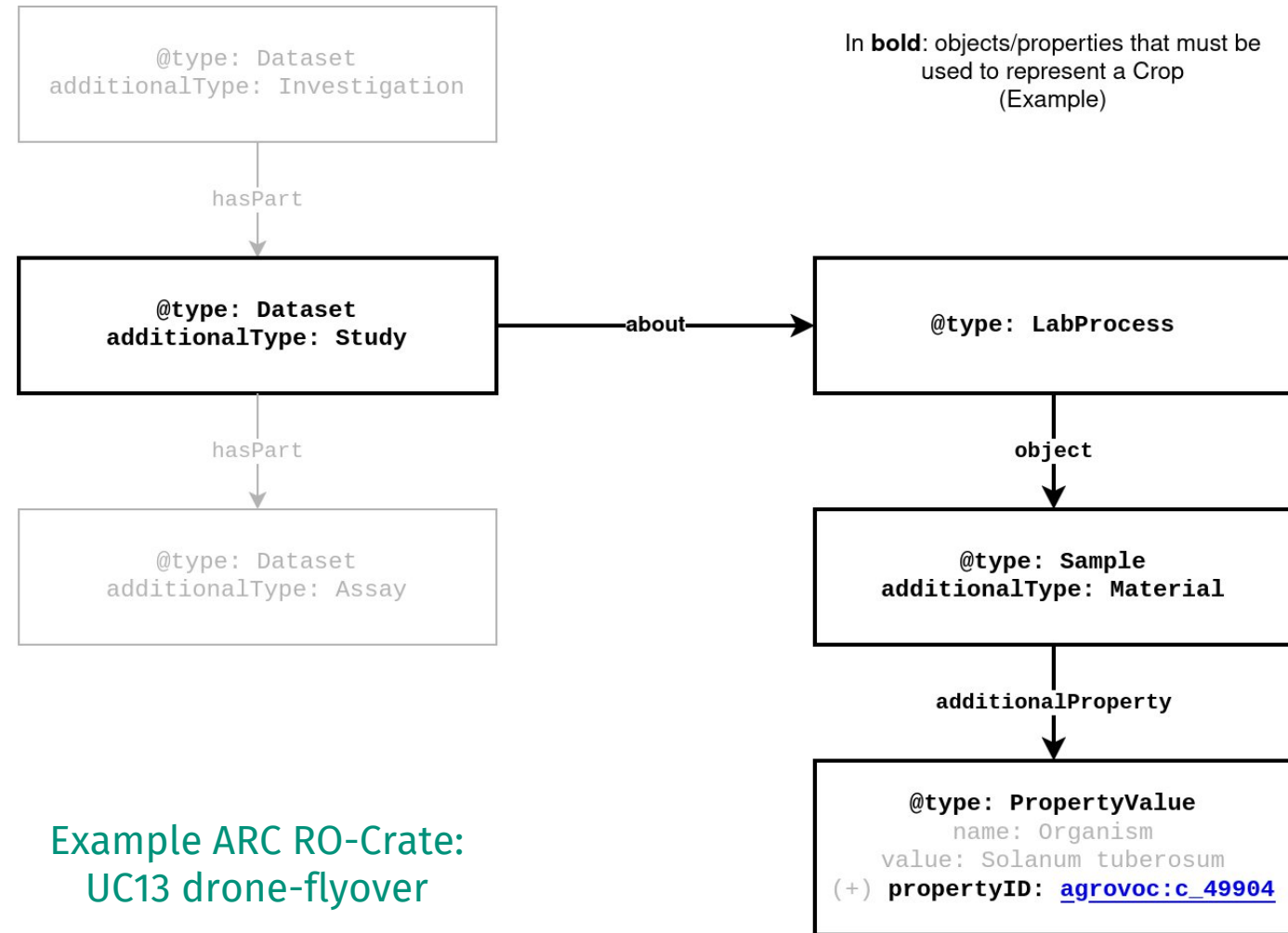
Example ARC RO-Crate:
UC13 drone-flyover

(Agrischemas: cropSpecies)

Examining ARC structure

Goal:

- define how Agrischemas-related objects should be modeled in ARCs
- enable unambiguous transformation



How to standardize these rules?

